PERFECT DIAGONAL FORCES SMOOTHNESS AT THE LOCAL LEVEL

THE STACKS PROJECT AUTHORS

Editorial note. Two standard identifications are left unverified in-span: the diagonal expansion ('a general fact about elements of A[x_1, ..., x_n] whose proof we omit') and 'Another computation shows that a_ij is the image of the partial derivative' – both are mechanical instantiations of the Jacobian presentation displayed two lines later; flagged for the reviewer.

Allowed standard prerequisites (editorial).

- Injectivity on the conormal level. lemma-injective (same chapter, named). Supplies I/mI -> J/mJ injective for the presentation data I subset J.
- Smoothness from the naive cotangent complex. algebra-lemma-smooth-at-point (cross-chapter, named). Turns the freeness of NL_{C/A} (x)_A B into an open neighbourhood smooth over A.

Source reference. Original reference 00DV: focused-seeds/geometry/S9-dpa-local-perfect-diagonal/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00HL: focused-seeds/geometry/S9-dpa-local-perfect-diagonal/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07BU: focused-seeds/geometry/S9-dpa-local-perfect-diagonal/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09PV: sources/stacks-dpa/pinned-original-source.tex, source lines 1790–1790. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 067H: focused-seeds/geometry/S9-dpa-local-perfect-diagonal/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 066E: focused-seeds/geometry/S9-dpa-local-perfect-diagonal/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $A \to B$ be a local ring homomorphism of Noetherian local rings such that B is flat and essentially of finite type over A. If*

$$B \otimes_A B \longrightarrow B$$

is a perfect ring map, i.e., if B has finite tor dimension over $B \otimes_A B$, then B is the localization of a smooth A-algebra.

Proof. As B is essentially of finite type over A, so is $B \otimes_A B$ and in particular $B \otimes_A B$ is Noetherian. Hence the quotient B of $B \otimes_A B$ is pseudo-coherent over $B \otimes_A B$ (More on Algebra, Lemma 066E) which explains why perfectness of the ring map (More on Algebra, Definition 067H) agrees with the condition of finite tor dimension.

We may write $B = R/K$ where R is the localization of $A[x_1, \ldots, x_n]$ at a prime ideal and $K \subset R$ is an ideal. Denote $\mathfrak{m} \subset R \otimes_A R$ the maximal ideal which is the inverse image of the maximal ideal of B via the surjection $R \otimes_A R \to B \otimes_A B \to B$. Then we have surjections

$$(R \otimes_A R)_{\mathfrak{m}} \to (B \otimes_A B)_{\mathfrak{m}} \to B$$

and hence ideals $I \subset J \subset (R \otimes_A R)_{\mathfrak{m}}$ as in Lemma 09PV. We conclude that $I/\mathfrak{m}I \to J/\mathfrak{m}J$ is injective.

Let $K = (f_1, \ldots, f_r)$ with r minimal. We may and do assume that $f_i \in R$ is the image of an element of $A[x_1, \ldots, x_n]$ which we also denote f_i. Observe that I is generated by $f_1 \otimes 1, \ldots, f_r \otimes 1$ and $1 \otimes f_1, \ldots, 1 \otimes f_r$. We claim that this is a minimal set of generators of I. Namely, if κ is the common residue field of R, B, $(R \otimes_A R)_{\mathfrak{m}}$, and $(B \otimes_A B)_{\mathfrak{m}}$ then we have a map $R \otimes_A R \to R \otimes_A \kappa \oplus \kappa \otimes_A R$ which factors through $(R \otimes_A R)_{\mathfrak{m}}$. Since B is flat over A and since we have the short exact sequence $0 \to K \to R \to B \to 0$ we see that $K \otimes_A \kappa \subset R \otimes_A \kappa$, see Algebra, Lemma 00HL. Thus restricting the map $(R \otimes_A R)_{\mathfrak{m}} \to R \otimes_A \kappa \oplus \kappa \otimes_A R$ to I we obtain a map

$$I \to K \otimes_A \kappa \oplus \kappa \otimes_A K \to K \otimes_B \kappa \oplus \kappa \otimes_B K.$$

The elements $f_1 \otimes 1, \ldots, f_r \otimes 1, 1 \otimes f_1, \ldots, 1 \otimes f_r$ map to a basis of the target of this map, since by Nakayama's lemma (Algebra, Lemma 00DV) $f_1, \ldots, f_r$ map to a basis of $K \otimes_B \kappa$. This proves our claim.

The ideal J is generated by $f_1 \otimes 1, \ldots, f_r \otimes 1$ and the elements $x_1 \otimes 1 - 1 \otimes x_1, \ldots, x_n \otimes 1 - 1 \otimes x_n$ (for the proof it suffices to see that these elements are contained in the ideal J). Now we can write

$$f_i \otimes 1 - 1 \otimes f_i = \sum g_{ij}(x_j \otimes 1 - 1 \otimes x_j)$$

for some g_{ij} in $(R \otimes_A R)_{\mathfrak{m}}$. This is a general fact about elements of $A[x_1, \ldots, x_n]$ whose proof we omit. Denote $a_{ij} \in \kappa$ the image of g_{ij}. Another computation shows

that a_{ij} is the image of $\partial f_i/\partial x_j$ in κ. The injectivity of $I/\mathfrak{m}I \to J/\mathfrak{m}J$ and the remarks made above forces the matrix (a_{ij}) to have maximal rank r. Set

$$C = A[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$$

and consider the naive cotangent complex $NL_{C/A} \cong (C^{\oplus r} \to C^{\oplus n})$ where the map is given by the matrix of partial derivatives. Thus $NL_{C/A} \otimes_A B$ is isomorphic to a free B-module of rank $n - r$ placed in degree 0. Hence C_g is smooth over A for some $g \in C$ mapping to a unit in B, see Algebra, Lemma 07BU. This finishes the proof. $\square$